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Paleomagnetic evidence for post-Cretaceous internal deformation of the Chuan Dian Fragment in the Yangtze block: a consequence of indentation of India into Asia

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Abstract

Paleomagnetic samples of Paleocene–Eocene red sandstones were collected at 36 sites from the Jiangdihe-4 and Zhaojiadian formations around the Yongren (26.1°N, 101.7°E) and Dayao areas (25.7°N, 101.3°E). These areas are located in the Chuxiong basin of the Chuan Dian Fragment, southwestern part of the Yangtze block. After stepwise thermal demagnetization, a high-temperature component with unblocking temperature of about 680 °C is isolated from 26 sites. The primary nature of this magnetization is ascertained through positive fold and reversal tests at 95% confidence level. The tilt-corrected mean paleomagnetic directions for the Yongren and Dayao areas are $D=17.2°$, $I=26.6°$ with $\alpha_{95}=5.8°$ and $D=16.5°$, $I=31.1°$ with $\alpha_{95}=4.8$, respectively. Easterly deflected declinations from this study are consistent with those reported from other areas of the Chuxiong basin, indicating its wide presence in the Cretaceous–Eocene formations of the said basin. Comparison with declination values expected from the Cretaceous–Eocene APWP of Eurasia indicates that the magnitude of clockwise rotation systematically increases toward the southeast within the Chuxiong basin as well as in the Chuan Dian Fragment. This trend of the differential tectonic rotation in the Chuan Dian Fragment is consistent with curvature of the Xianshuihe–Xiojiang fault system. Deformation of the Chuxiong basin can fairly be associated with the formation of eastward bulge in the southern part of the Chuan Dian fragment. During southward displacement, the Chuan Dian Fragment was probably subjected to tectonic stresses as a result interaction with the Yangtze and Indochina blocks, which resulted into east–west extension and north–south shortening.
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Keywords: Tectonics; Continent; Chuan Dian Fragment; Paleomagnetism; Early Tertiary

1. Introduction

The crust of East Asia comprises several continental blocks which are bounded by lithospheric scale faults. These faults played a significant role in east–southeastward extrusion of tectonic blocks since the time of India–Asia collision at about 50 Ma (e.g. Tapponnier et al., 1982; England and Molnar, 1990, Avouac and Tapponnier, 1993; Lacassin et al., 1997). The Chuan Dian Fragment, which is one of prominent examples of tectonic displacement (Fig. 1), occupies

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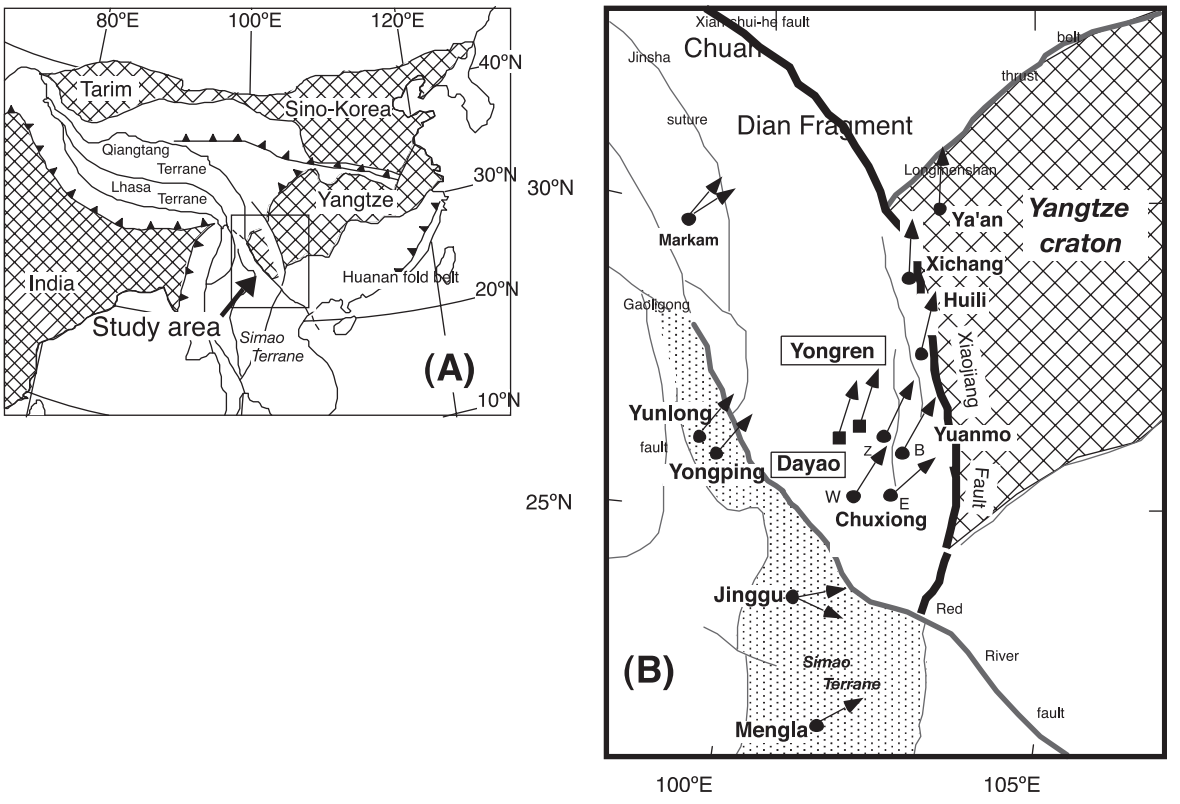


Fig. 1. (A) Tectonic map of the Southeast Asia. Black and barbed lines indicate suture zones and thrusts, respectively. (B) Schematic map of the study area, showing different areas of the Yangtze block, the Chuan Dian Fragment and the Simao Terrane. Solid squares are sampling localities for this study. Solid circles indicate the areas where Cretaceous paleomagnetic results are available (Otofuji et al., 1990, 1998; Enkin et al., 1991a,b; Huang and Opdyke, 1992, 1993; Huang et al., 1992; Funahara et al., 1992; 1993; Chen et al., 1995; Sato et al., 1999; Yang et al., 2001). Cretaceous to Eocene declination values are indicated by arrows. Z, Zhupeng; B, Bailu; W, the Chuxiong west area; E, the Chuxiong east area.

the southwestern part of the Yangtze block. The Xianshuihe–Xiaojiang fault system separates it from the main part of the Yangtze block. This fragment is bounded to the southwest by the Shan–Thai–Indochina block (Li et al., 1982) and the boundary is the NW–SE-trending Red River fault, which is connected with the Jinsha suture toward north in Tibet. GPS measurements and geological observations along the Xianshuihe–Xiojiang fault system suggest a southward displacement of the Chuan Dian Fragment during the last 12 m.y. (Roger et al., 1995; Wang et al., 1998, 2001; Chen et al., 2000). Previous paleomagnetic studies reported from the area suggest that this fragment experienced more than 300 km of southward displacement after Cretaceous times (Yokoyama et al., 1999; Tamai et al., 2003).

The indentation of India into Asia brought about some internal deformation to the Chuan Dian Fragment during its southward translation. This deformation can be seen in significant amount of rotation deduced from Cretaceous paleomagnetic data of this fragment (Fig. 1). The available paleomagnetic data indicate that declination values increase latitudinally from north to south (Funahara et al., 1992; Huang and Opdyke, 1992; Otofuji et al., 1998; Tamai et al., 2003) and the magnitude of rotational motion significantly increases in the southern part of this fragment.

To better understand the tectonics-related internal deformation of the Chuan Dian Fragment, we collected paleomagnetic samples from the Chuxiong basin, the southern part of this fragment. The Yongren and Dayao areas of this basin were chosen for present

paleomagnetic study to detect longitudinal variation of paleomagnetic direction in the Chuxiong basin.

2. Geological setting and sampling

The Chuxiong basin is located in the southeastern part of the Chuan Dian Fragment (Fig. 1). This basin is bounded by the Chuxiong fault to the south and the Yuanmou fault to the east. The Chuxiong fault had a left-slip displacement in the middle Cenozoic, but recent earthquake focal mechanism study indicates an active right-slip movement on this fault (Wang et al., 1998). The Yuanmou fault generally constitutes a N–S strike with unsymmetrical horst.

A detailed geology of the Chuxiong basin can be seen in the reports published by Bureau of Geology and Mineral Resources of Yunnan Province (1990) and Leloup et al. (1995). In this basin, Upper Triassic gray sandstones and marls grade upwards into Jurassic and Cretaceous red sandstones. The youngest redbeds of this basin are placed in Eocene age. There is no clear evidence of definite regional unconformity within this Mesozoic–Tertiary sequence. The Upper Cretaceous–Paleocene series is called the Jiangdihe-1 to Jiangdihe-4 formations, which are 2574 m thick, and made up of purple-grayish green sandy mudstone and shale, containing elements of *Cypridea* (*Ulwellia*) houmafangenesis and *Manchurichthys* sp. The Jiangdihe-3 and Jiangdihe-4 formations are assigned to Paleocene by Bureau of Geology and Mineral Resources of Yunnan Province (1990). The Eocene series of the Zhaojiadian Formation, which is 1151 m thick and conformably overlies the Paleocene Jiangdihe-4 Formation, is made up of purple-red sandstone and mudstone, containing elements of *Carpolithus* sp. and *Ginkgo* sp.

The Upper Cretaceous–Eocene redbeds in the Chuxiong basin is folded into NW to NNW trending anticlines (Leloup et al., 1995). The folding took place in the Upper Eocene–Oligocene, as evidenced by an unconformity between the Lower Eocene and the Upper Eocene–Oligocene (Bureau of Geology and Mineral Resources of Yunnan Province, 1990).

We sampled the Paleocene–Eocene red sandstones of the Jiangdihe-4 and Zhaojiadian formations for paleomagnetic study (Fig. 2). Samples were collected from two basins, i.e., 23 sites in the Yongren area (26.05°N, 101.67°E) and 13 sites in the Dayao area (25.73°N, 101.32°E). Sampling sites are distributed on both limbs of the synclinal structure in each basin. Eight to ten hand samples, oriented with a magnetic compass, were collected over a distance of up to 15 m from each site. Location coordinates of the sampling sites were determined by a portable navigational instrument (SONY PYXIS). The present declination value at each sampling locality was evaluated from the International Geomagnetic Reference Field, 1995 (IAGA Division V, Working Group 8, 1995).

3. Laboratory procedure

In the laboratory, one specimen of 25 mm in diameter and 23 mm in height were prepared from each sample. Natural remanent magnetization (NRM) of each specimen was measured using a 2G Enterprises cryogenic magnetometer (SQUID). Stepwise thermal demagnetization was carried out with a laboratory-built furnace or a Natsuhara TDS-1 thermal demagnetizer with an internal residual filed of less than 5 nT. Each specimen was subjected to 17 heating steps up to 690 °C with temperature intervals of 10 °C above 650 °C.

Remanent magnetic vectors were plotted on orthogonal diagrams (Zijderveld, 1967) to assess component structure as well as on equal-area projections to evaluate directional stability. Principal component analysis (Kirschvink, 1980) was used to determine the directions of different magnetization component. Mean direction was calculated using Fisherian statistics (Fisher, 1953).

4. Paleomagnetic results

4.1. Yongren area

Initial NRM intensity of the Zhaojiadian formation ranges between 0.19×10^{-8} and 12.0×10^{-8} A m^2. Specimens from the Jiangdihe-4 Formation gave relatively stronger intensities ranging between 0.37×10^{-8} and 16.0×10^{-8} A m^2. Almost all specimens from both formations revealed two components of magnetization (Fig. 3). The low-temperature component is generally isolated by 350 °C. The high-

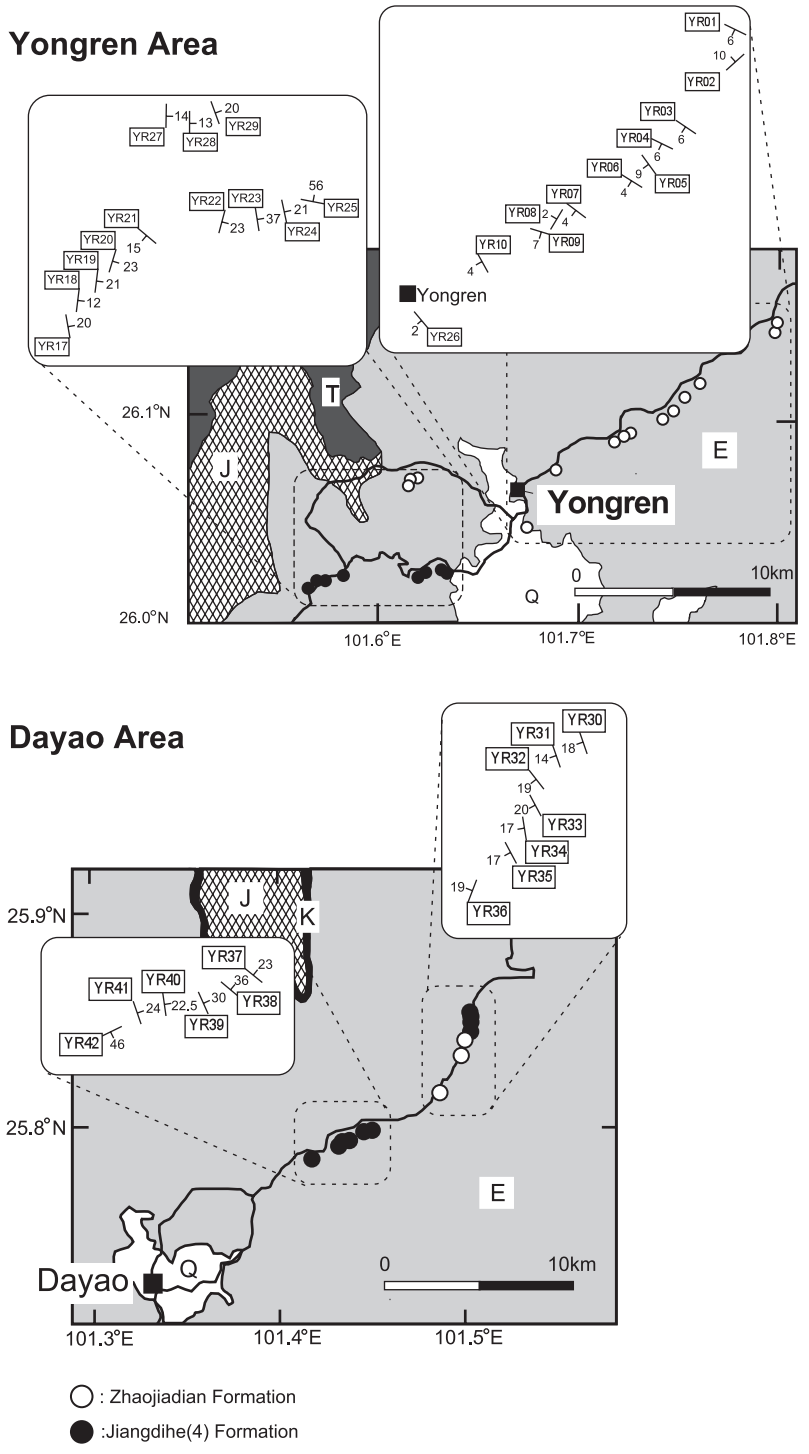


Fig. 2. Geological maps of the study areas. Sampling sites are shown by open and closed circles for the Zhaojiadian and Jiangdihe-4 formations, respectively. Strike/dip orientations of strata are shown in insets. T, Triassic; J, Jurassic; K Cretaceous; E, Paleocene to Eocene, Q, Quaternary.

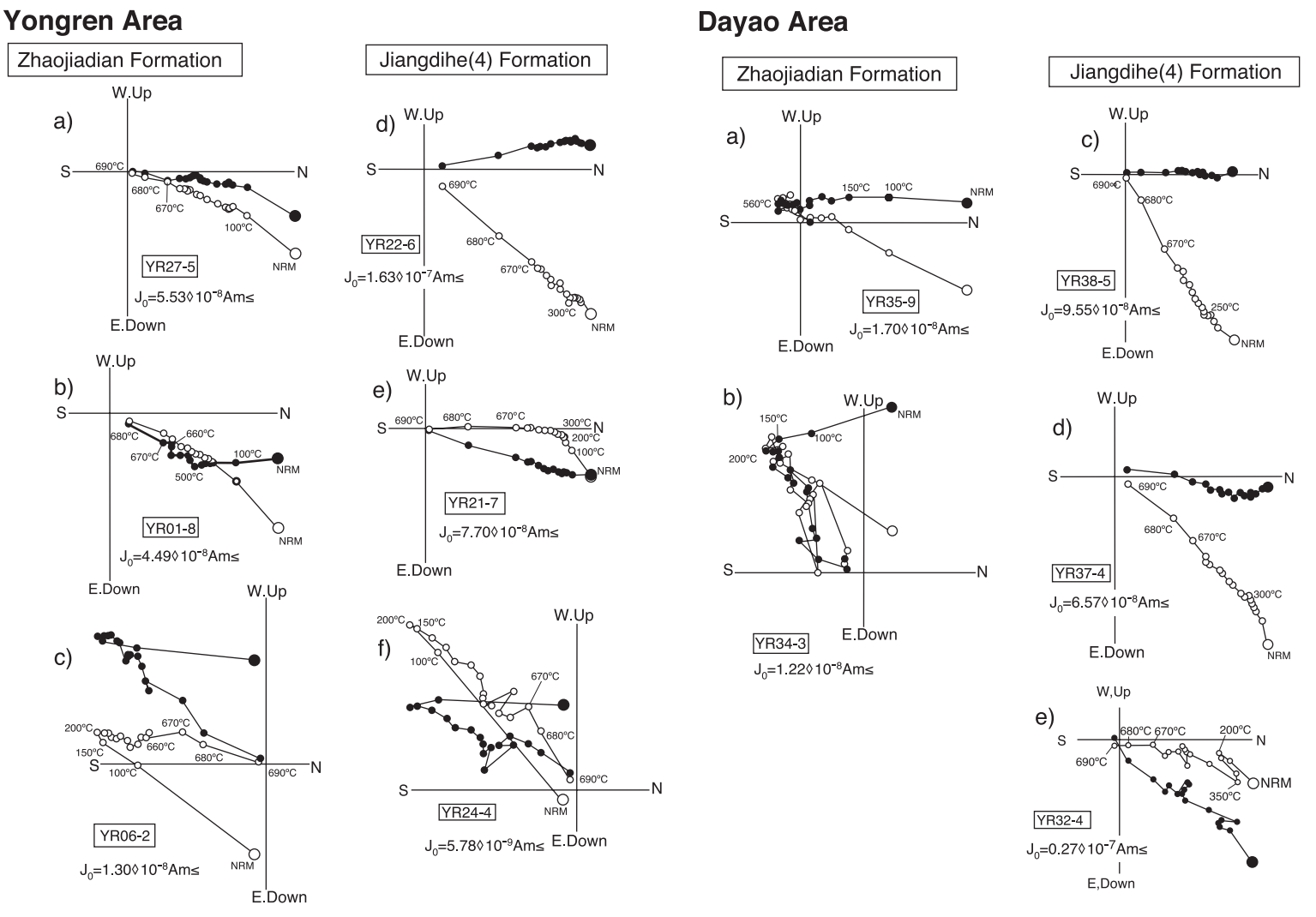


Fig. 3. Orthogonal projections of magnetization in geographic coordinates after thermal treatments for the Yongren and Dayao areas. Open and solid symbols show projection onto the vertical and the horizontal planes, respectively. Site name (YR27) and specimen name (5) are shown as YR27-5, for instance.

Table 1
Paleomagnetic results of the Paleocene–Eocene redbeds from the Yongren and Dayao areas

Site	Locality		*n*/*N*	Polarity	In situ		Tilt-corrected		k	α_{95}	Bedding
	°N	°E			D (deg)	I (deg)	D (deg)	I (deg)		(deg)	strike (deg)/ dip (deg)
Yongren Area											
Zhaojiadian Formation (Eocene)											
East wing											
YR01	26.15	101.80	7/9	N	23.4	30.8	23.3	36.8	20.6	13.6	116/6
YR02	26.14	101.80	7/8	N	25.1	22.8	21.6	18.8	32.1	10.8	228/10
YR03*	26.12	101.76	5/9	I	231.0	−24.0	231.9	−29.6	4.7	39.7	124/6
YR04*	26.11	101.75	4/9	I	301.4	15.4	299.7	15.8	7.5	36.0	116/6
YR05	26.10	101.75	0/9	–	–	–	–	–	–	–	114/9
YR06	26.10	101.74	5/9	R	197.3	−19.1	197.0	−23.0	26.7	15.1	122/4
YR07	26.09	101.73	4/8	R	206.0	−37.7	205.4	−41.6	17.0	22.9	127/4
YR08	26.09	101.72	0/9	–	–	–	–	–	–	–	211/2
YR09*	26.09	101.72	4/8	R	212.2	−11.8	212.7	−18.6	7.1	37.1	107/7
YR10	26.07	101.69	4/8	R	204.0	−17.2	203.2	−20.5	10.2	30.3	151/4
YR26*	26.04	101.67	4/9	R	152.8	−23.2	152.9	−22.7	6.2	40.1	320/20
West wing											
YR27	26.07	101.61	5/9	N	359.6	19.4	6.2	18.4	333.1	4.2	1/19
YR28	26.07	101.61	7/8	R	176.7	−18.4	180.0	−18.5	37.9	9.9	0/13
YR29	26.07	101.62	5/9	R	171.8	−28.9	181.2	−22.8	24.4	15.8	340/20
Jiangdihe-4 Formation (Paleocene)											
YR17	26.01	101.56	9/9	N	4.7	39.0	17.9	31.3	85.5	5.6	350/20
YR18	26.02	101.57	9/9	N	7.6	32.6	15.1	31.5	67.7	6.3	7/12
YR19	26.02	101.57	9/9	N	1.1	31.7	14.1	31.1	84.9	5.6	7/12
YR20	26.02	101.57	9/9	N	349.9	35.1	7.8	42.1	38.4	8.4	17/23
YR21	26.02	101.58	9/9	N	25.4	8.1	24.3	22.6	26.0	10.3	130/15
YR22	26.02	101.62	9/9	N	3.9	32.5	19.1	33.5	52.2	7.2	15/23
YR23	26.02	101.62	5/9	N	9.7	34.0	25.7	16.2	20.7	17.2	351/37
YR24	26.02	101.63	4/9	R	207.7	−27.2	213.3	−12.6	16.5	23.3	348/21
YR25*	26.02	101.63	8/9	I	52.5	0.6	68.1	−37.4	6.4	23.8	281/56
Mean direction											
All	26.10	101.70	16/23		11.5	27.7			32.3	6.6	
							17.2	26.6	42.1	5.8	
Normal			10/16		9.4	29.1			33.6	8.5	
							17.7	29.4	59.3	6.3	
Reverse			6/16		193.8	−25.4			25.1	13.6	
							196.5	−23.6	26.2	13.3	
Dayao Area											
Zhaojiadian Formation (Eocene)											
YR34	25.84	101.50	0/9	–	–	–	–	–	–	–	171/17
YR35	25.83	101.49	0/9	–	–	–	–	–	–	–	152/17
YR36	25.81	101.48	0/9	–	–	–	–	–	–	–	201/19
Jiangdihe-4 Formation (Paleocene)											
YR30	25.85	101.50	8/9	N	10.5	27.7	1.5	35.7	54.6	7.6	161/18
YR31	25.85	101.50	6/9	N	26.2	16.8	22.1	26.6	21.7	14.7	161/14
YR32	25.85	101.50	8/9	N	26.9	14.0	23.4	31.2	27.3	10.8	142/19
YR33	25.84	101.50	8/9	N	25.0	13.4	20.1	29.1	17.3	13.7	152/20
YR37	25.79	101.45	8/9	N	11.2	42.4	17.0	21.4	33.2	9.8	309/23
YR38	25.79	101.44	9/9	N	356.0	55.1	13.6	24.7	151.4	4.2	310/36
YR39	25.79	101.44	9/9	N	357.2	52.7	21.6	35.1	58.5	6.8	336/30
YR40	25.79	101.43	5/8	N	349.9	37.1	8.1	33.5	30.3	14.1	352/26
YR41	25.79	101.43	9/10	N	1.6	44.4	18.3	32.2	60.9	6.6	341/24

Table 1 (*continued*)

Site	Locality		*n*/*N*	Polarity	In situ		Tilt-corrected		k	α_{95}	Bedding
	°N	°E			D (deg)	I (deg)	D (deg)	I (deg)		(deg)	strike (deg)/ dip (deg)
Dayao Area											
Jiangdihe-4 Formation (Paleocene)											
YR42	25.78	101.42	9/9	N	5.9	3.3	18.1	40.1	56.4	6.9	64/46
Mean direction											
All	25.70	101.30	10/13		10.7	31.3			14.5	13.1	
							16.5	31.1	100.4	4.8	

Magnetization directions of high-temperature component are listed. N, R and I represent normal, reversed and intermediate polarity directions, respectively. *N* (*n*) is number of samples measured (number of samples used for calculation); D and I are declination and inclination, respectively; *k* is the estimate of the precision parameter (Fisher, 1953); α_{95} and A_{95} are the radii of the cone of 95 % confidence. Strikes of bedding are values before the subtraction of local geomagnetic declination. A value of 359.1° is calculated for the present local geomagnetic declination using the Definitive Geomagnetic Reference Field of 1995 (IAGA Division V, Working Group 8, 1995). Data sites denoted by asterisk are excluded from formation mean calculation because of low precision parameter values ($k<10$).

temperature component is unblocked between 680 and 690 °C, indicating hematite as a carrier of magnetization (Table 1).

For both formations, the low-temperature component is isolated from 18 sites. The mean direction of this components is $D=358.4°$, $I=38.8°$ with $\alpha_{95}=7.3°$ in situ and $D=3.4°$, $I=37.4°$ with $\alpha_{95}=12.3°$ after tilt correction. This mean direction passes a negative fold test at 99% confidence level (McElhinny, 1964), indicating a post-folding origin for low-temperature component. Since the in situ mean direction is hardly distinguishable from the present Earth's magnetic field direction at the Yongren area ($D=359.1°$, $I=38.6°$), this component is ascribed to be a viscous remanent magnetization acquired in recent times.

The high-temperature component is identified in 21 sites. Both normal and reversed polarities are observed in sites from both the formations (Fig. 4). The site mean directions for this component in 16 out of 21 sites show tight clustering, where precision parameter ranges from 10.2 to 333.1. Since the remaining five sites (YR03, YR04, YR09, YR25, YR26) show low precision parameter ($k<10$), these data are discarded for calculation of the formation mean direction.

Tilt-corrected mean direction of the high-temperature component calculated for the Zhaojiadian Formation is $D=14.3°$, $I=25.3°$, $\alpha_{95}=9.1°$, $N=8$ and that for the Jiangdihe-4 Formation is $D=20.2°$, $I=27.8°$, $\alpha_{95}=8.3°$, $N=8$. These two formation mean directions are indistinguishable at 95% confidence level, where the difference in direction is only 5.6°. We combined all the 16 sites from both formations to estimate a characteristic direction of the Paleocene–Eocene redbeds at the Yongren area. The combined mean direction is $D=11.5°$, $I=27.7°$, with $\alpha_{95}=6.6°$ (in situ), and $D=17.2°$, $I=26.6°$, with$\alpha_{95}=5.8°$ (after tilt correction). Although the fold test of McElhinny (1964) is inconclusive at 95% confidence level ($k_s/k_g=1.30$), rigorous fold tests proposed by McFadden and Jones (1981) and McFadden (1990) are positive at 95% confidence level. The *p* value from McFadden and Jones fold test is 0.3527 in geographic coordinates and 0.0655 in stratigraphic coordinates, while the critical value is 0.2386 at 95% confidence level. In the McFadden's fold test, the calculated value (ξ_1) is 4.672 in situ coordinates and 0.355 after tilt correction, while the critical value (ξ) is 4.658 at the 95% confidence level. These positive fold tests suggest a pre-folding origin for the high temperature component.

The reversal test is also positive at 95% confidence level (McFadden and McElhinny, 1990). The γ_c is 12.1° from the data set of normal and reversed directions, while difference between normal and antipodal reversed mean directions is 4.9°, indicating that a reversal test is passed at 95% confidence level as 'C' classification. The presence of dual polarity indicates that the mean paleomagnetic direction represents a significant time average in Paleocene–Eocene formation.

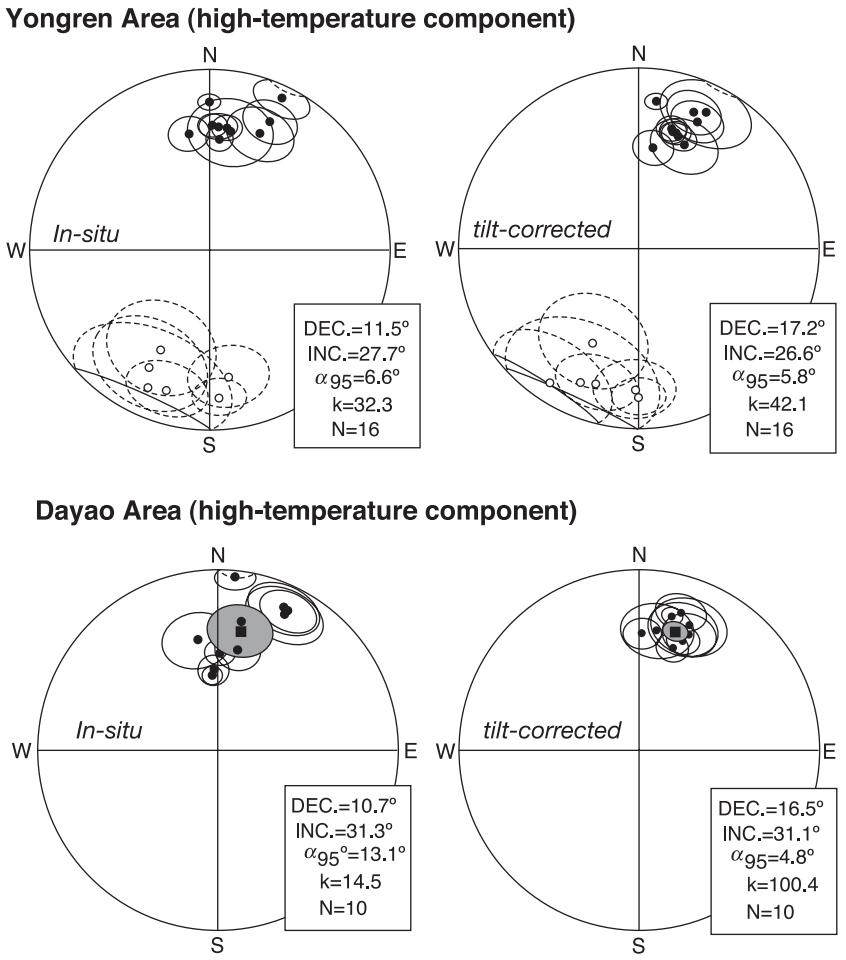


Fig. 4. Equal-area projections of site mean directions with 95% confidence circles for the Yongren and Dayao areas. Solid (open) symbols are lower (upper) hemisphere projection. Formation mean is shown by square.

4.2. Dayao area

As observed in Yongren area, initial NRM intensities of the Zhaojiadian Formation (0.57 to 1.82×10^{-8} A m^2) is weaker than that of the Jiangdihe-4 Formation (1.02 to 18.0×10^{-8} A m^2). Because a specimen of the Zhaojiadian Formation is made up of sandstone, content of magnetic mineral in it is presumably smaller than sandy mudstone of the Jiangdihe-4 Formation. All specimens from the Zhaojiadian formation preserve the low-temperature component, which is generally erased by 350 °C. But no high-temperature component is isolated by the principal component analysis because spurious remanent magnetization made it difficult above this temperature (Fig. 3).

Almost all specimens from the Jiangdihe-4 Formation reveal two or three components of magnetization (Fig. 3). The low-temperature component is erased by 350 °C. The intermediate-temperature component, which appeared only in 14 specimens, is separated between 250 and 350 °C. The high-temperature component is unblocked between 680 and 690 °C, indicating hematite as a carrier of magnetization.

The low-temperature component is isolated in 10 out of 13 sites of both the formations. The in situ mean direction ($D = 350.6°$, $I = 41.4°$ with $\alpha_{95} = 10.0°$)

is nearly identical to the present Earth's magnetic field direction at the Dayao area ($D=359.2°$, $I=38.0°$), indicating a recent origin for this component.

A mean direction of the intermediate-temperature component is $D=199.3°$, $I=-39.6°$ with $\alpha_{95}=9.5°$ (in situ), and $D=185.4°$, $I=-52.1°$ with $\alpha_{95}=9.6°$ (after tilt correction). This component was probably acquired before the Brunhes chron because the mean direction is parallel to the antipodal of present Earth's field direction within 95% confidence level. Unblocking temperature of 350 °C implies pyrrhotite and/or maghemite as remanence carriers for this component.

The high-temperature component obtained from 10 sites of the Jiangdihe-4 Formation show tight clustering within site. The precision parameters from these site mean directions range from 17.3 to 151.4. As shown in Fig. 4, the formation mean direction is $D=10.7°$, $I=31.3°$, with $\alpha_{95}=13.1°$ (in situ), and $D=16.5°$, $I=31.1°$, with $\alpha_{95}=4.8°$ (after tilt correction). The precision parameter (k) increases from 14.5 to 100.4 ($k_s/k_g=6.92$) after tilt correction. Applying McElhinny (1964) and McFadden (1990) fold tests, positive results are obtained at the 99% confidence level. The positive fold tests clearly indicate that the high-temperature component in the samples of the Jiangdihe-4 Formation is pre-folding in origin.

5. Discussion and interpretation

5.1. Change of declination in the Chuxiong basin

The characteristic remanent magnetization obtained from the high-temperature component gave a tilt-corrected mean direction of $D=17.2°$, $I=26.6°$, $\alpha_{95}=5.8°$ for Yongren area and $D=16.5°$, $I=31.3°$, $\alpha_{95}=4.8°$ for Dayao area. Pre-folding nature is ascertained through positive fold test for this component obtained from the redbeds of the Zhaojiadian and Jiangdihe-4 formations. This magnetization is probably primary in origin because folding in the Yongren and Dayao areas took place in Late Eocene–Oligocene just after the formation of Paleocene–Eocene redbeds (Bureau of Geology and Mineral Resources of Yunnan Province, 1990). The easterly deflected direction obtained from both areas represents a record with a fairly long period because of the positive reversal test.

Previously, an easterly paleomagnetic direction has been reported through different paleomagnetic studies from the Chuxiong basin (Fig. 5). Paleocene paleomagnetic directions of the Jiangdihe-3 Formation from Zhupeng and Bailu localities in the Yuanmou area are $D=25.4°$, $I=34.3°$, $\alpha_{95}=3.4°$ and $D=30.7°$, $I=39.0°$, $\alpha_{95}=10.6°$, respectively (Otofuji et al., 1998). An easterly declination ($D=42.9°$, $I=43.3°$, $\alpha_{95}=11.0°$) is also reported from the Upper and Lower Cretaceous redbeds of the southern part of the Chuxiong basin (Funahara et al., 1992). The easterly deflected declination thus characterizes the trend of paleomagnetic directions in the Cretaceous to Eocene strata of the Chuxiong basin, the southern part of the Chuan Dian Fragment. However, these easterly deflected directions are not in agreement with the northerly declination reported from Huili and Xichang areas of the northern part of the Chuan Dian Fragment (Huang and Opdyke, 1992; Tamai et al., 2003) (Fig. 1).

Our new discovery is longitudinal increase toward the east in change of declination values (Fig. 5). Declination data from the northern part of the Chuxiong basin (including the Yongren, Dayao, Zhupen and Bail areas) are plotted as a function of longitude for sampling sites (Fig. 5B). The least square fitting method indicates that declination increases from 14° to 32° in a range of longitude from 101.4°E to 102.0°E. With respect to the declination of the Dayao area, an increase of 0.7°, 8.9° and 14.2° is indicated for the Yongren, Zhupeng and Bailu areas, respectively.

An easterly increase in declination value can also be observed in the Cretaceous paleomagnetic data (21 sites) reported from the southern part of the Chuxiong basin (Funahara et al., 1992). We selected reliable data from only 12 out of these 21 sites (numbered in Fig. 5A) to detect this aspect precisely. Data of seven sites were discarded because of their location within 3 km of the fault, since crustal rotation due to fault motion is expected within a few kilometers from a fault zone (Otofuji et al., 1995; Piper et al., 1997). Two reversed polarity directions were also excluded because of their exceptionally large westerly deflection (site 76: $D=295.6°$, site 88: $D=253.6°$), which may be due to a reversal or excursion of the geomagnetic field.

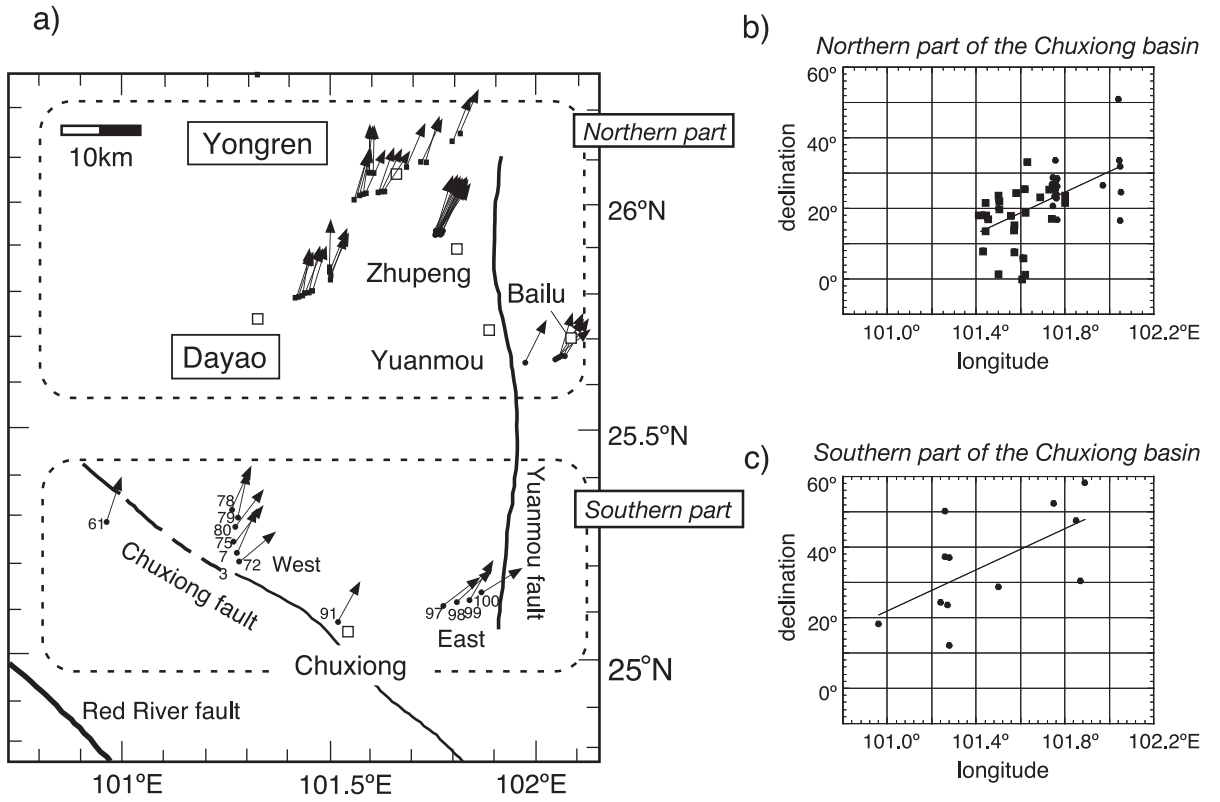


Fig. 5. Internal rotation of paleomagnetic directions in the Chuxiong basin. (A) Cretaceous declination data from the Chuxiong basin. Sampling sites for Yongren and Dayao areas of this study are shown by solid squares. Solid circles indicate the areas of previous Cretaceous paleomagnetic results (Funahara et al., 1992; Otofuji et al., 1998). Paleomagnetic declination values are indicated by arrows. (B and C) Declination data from the northern and southern parts of the Chuxiong basin are plotted as a function of longitudes for each sampling site, respectively. Solid lines are fitted by least square method. Yongren, Dayao, Zhupeng and Bailu areas constitute the northern part of the Chuxiong basin, while the Chuxiong area is located in the southern part. Solid squares are data from Yongren and Dayao areas.

The selected 12 sites clearly show an increase in declinations toward the east (Fig. 5C). Using data from 10 out of 12 selected sites, mean paleomagnetic declinations are calculated for the west (W) and east (E) areas of the Chuxiong basin (the Chuxiong W consists of six sites, i.e., 72, 73, 75, 78, 79, 80, while the Chuxiong E consists of four sites, i.e., 97, 98, 99, 100) as $D=30.6°$ and $D=47.7°$, respectively. This type of behaviour indicates that longitudinal increase in declination is one of characteristic features in the Chuxiong basin (Fig. 5B and C).

Latitudinal increase in declination toward south is another characteristic feature observed in the Chuxiong basin, which is clearly manifested in a plot of declination versus latitude (Fig. 6). Declination values from the southern areas of the Chuxiong basin are larger than those of the northern areas (e.g., $D=47.8°$ for the Chuxiong E area vs. $D=26.8°$ for Bailu). According to the present study, a southward increase in easterly declination, which has been previously reported from other parts of the Chuan Dian Fragment (Otofuji et al., 1998), is also observed within the Chuxiong basin.

The declination features thus suggest that the Chuxiong basin was subjected to some sort of internal tectonic deformation. Variation in the declination values can generally be explained by the following factors: (1) polar wandering, (2) geomagnetic behaviour (e.g. secular variation and excursion) or (3) tectonic rotation. Since little change is reported during Late Cretaceous–Early Tertiary in the Eurasian APWP (Besse and Courtillot, 1991), variable declina-

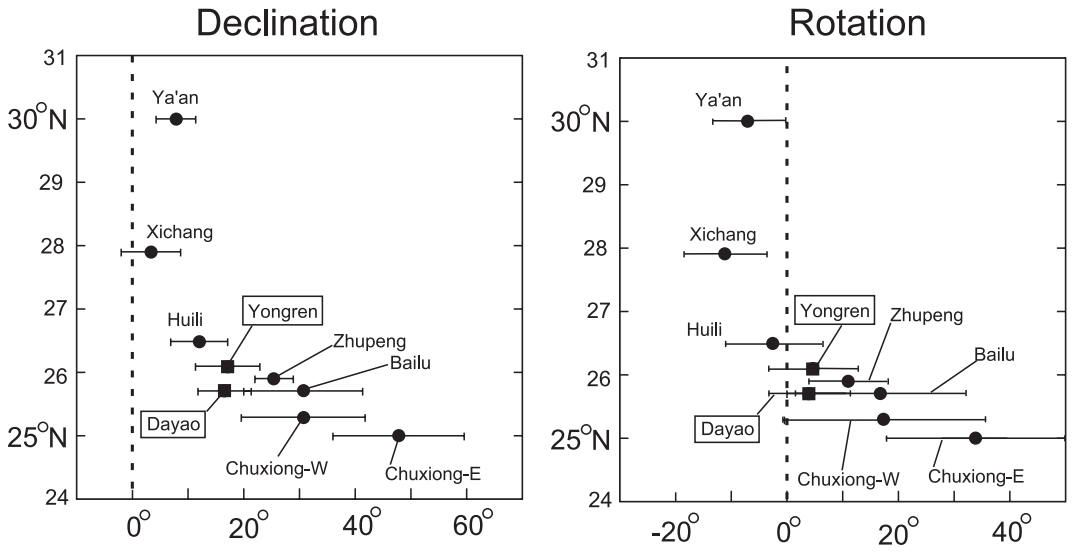


Fig. 6. Declination and rotation as a function of latitude in the Chuan Dian Fragment. Solid squares are data from present study. Solid circles represent the areas where Cretaceous paleomagnetic results are available (Enkin et al., 1991a,b; Funahara et al., 1992; Huang and Opdyke, 1992; Otofuji et al., 1998; Tamai et al., 2003). Tectonic rotation is estimated with respect to the stable part of the SCB (Besse and Courtillot, 1991).

tions from the Chuxiong basin cannot be attributed to this factor. The effects of secular variation or excursion are also discarded because the presence of dual polarities indicates a significant time average of the paleomagnetic direction. The most reasonable explanation is thus the effects of tectonic deformation,

Table 2
Cretaceous to Paleogene paleomagnetic directions and corresponding poles from the Chuan Dian Fragment and Sichuan basin

Locality			Age	Observed direction				Pole Position			Rotation $R \pm \Delta R$	Reference
Name	Lat. (°N)	Long. (°E)		N	D (deg)	I (deg)	α_{95} (deg)	Lat. (°N)	Long. (°E)	A_{95} (deg)		
Sichuan basin (Yangtze craton)												
Ya'an (Feixian Guan)	30.0	103	J–E	44	7.9	34.4	3.6	76.8	248.3	2.4/4.1	-7.2 ± 6.4^{b}	(Enkin et al., 1991b)
Chuan Dian Fragment												
Xichang	27.9	102.3	K1	33	3.7	41.5	3.4	85.2	241.7	3.5	-11.1 ± 7.2^{b}	(Tamai et al., 2003)
Huili	26.5	102.4	K1, K2	25	12.1	38.4	5.1	78.0	212.8	5.5	-2.5 ± 8.6^{b}	(Huang and Opdyke, 1992)
Chuxiong basin												
Yongren	26.1	101.7	E	16	17.2	26.6	5.8	70.1	224.6	4.9	4.5 ± 7.8^{a}	this study
Dayao	25.7	101.3	E	10	16.5	31.1	4.8	72.3	218.4	4.5	3.8 ± 7.1^{a}	this study
Zhupeng	25.9	101.8	K2, E	15	25.4	34.3	3.4	65.6	203.0	2.6	10.9 ± 6.9^{b}	(Otofuji et al., 1998)
Bailu	25.7	102.1	K2, E	6	30.7	39.0	10.6	61.8	192.2	10.5	16.2 ± 14.8^{b}	(Otofuji et al., 1998)
Chuxiong W	25.3	101.3	K1	6	30.6	48.5	11.1	62.5	173.9	12.2	16.9 ± 17.7^{b}	(Funahara et al., 1992)
Chuxiong E	25.3	101.8	K1	4	47.8	36.5	11.8	46.3	187.9	13.2	33.3 ± 15.8^{b}	(Funahara et al., 1992)

N is number of sites used for calculation. Tectonic rotation is estimated with respect to the stable part of the SCB (Besse and Courtillot, 1991). [a, b, c]: rotation with respect to the declination expected from the poles of 60, 100 and 120 Ma, respectively. The amount of rotation (R) and its uncertainty (ΔR) are calculated using the method of Magill et al. (1981). J, Jurassic; K1, Early Cretaceous; K2, Late Cretaceous; E, Paleocene to Eocene. For other explanation, see Table 1.

which paved a way for differential rotation in the Chuxiong basin.

5.2. Tectonic rotation and deformation of the Chuan Dian Fragment

Using a combination of the here presented and previously reported paleomagnetic data (Fig. 6 and Table 2), a tectonic rotation of the Chuan Dian Fragment is estimated with respect to the stable part of the South China Block (SCB). Since the Cretaceous paleomagnetic pole reported by Yang and Besse (2001) from stable SCB is compatible with the 100-Ma pole of Eurasia (Besse and Courtillot, 1991), we used the later one as reference for comparative statistics. From north to south in the Chuan Dian Fragment (Fig. 6), sense and magnitude of tectonic rotation systematically changes. Counterclockwise rotation of 11.1° is observed in the northern part of the Chuan Dian Fragment (the Xichang area), and statistically null rotation occurred at the Huili, Yongren and Dayao areas, whereas the southeastern part of the fragment experienced a clockwise rotation. Magnitude of clockwise rotation increases up to 33° at the Chuxiong E area.

Based on the input data points of eight locations shown as closed circles in Fig. 7A (Xichang, Huili, Yongren, Dayao, Zhupen, Bailu, Chuxiong W and Chuxiong E), we applied a solution using a tension parameter $T_I = 1$ (Smith and Wessel, 1990; Wessel and

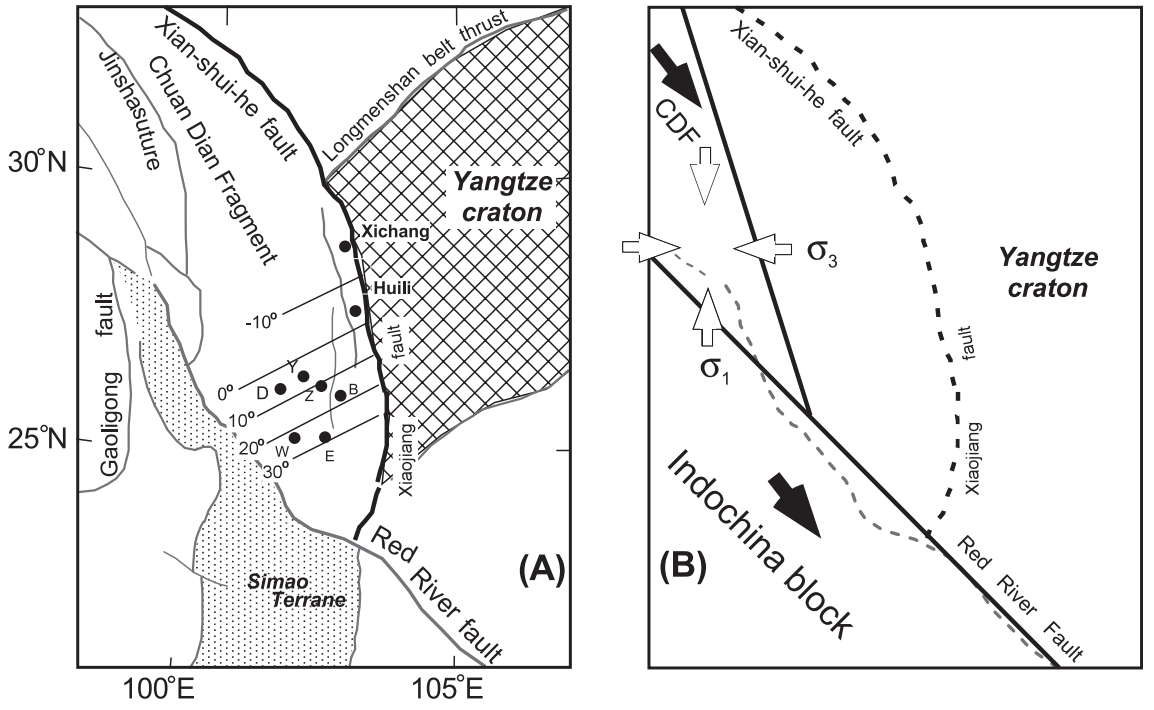


Fig. 7. (A) Contours of the amount of rotation in the southern part of the Chuan Dian Fragment. Contours at 10° interval are calculated from the input data values of eight localities shown as closed circles [Xichang, Huili, Yongren (Y), Dayao (D), Zhupeng (Z), Bailu (B), Chuxiong W (W) and Chuxiong E (E)]. For this purpose, General Mapping Tools of Wessel and Smith (1991) are used. (B) A deformation model for the Chuan Dian Fragment. The paleoshape of the Chuan Dian fragment was originally apex. An initially linear Xianshuihe–Xiojian fault system (solid line) changed into present curvilinear shape (dashed line) as a result of deformation in the Chuan Dian Fragment. During its southward displacement (closed arrows) in the Cenozoic era, the Chuan Dian Fragment tectonically interacted with the Yangtze and Indochina blocks. Roughly, east–west extension (σ_1) and north–south compression (σ_3) are raised up (open arrows) in the southern part of the Chuan Dian Fragment. Due to these compressive stresses, an eastward bulge has been formed in the Xianshuihe–Xiojian fault system accompanied by a differential tectonic rotation in the Chuxiong basin.

Smith, 1991) to depict first-order approximation of rotation in the Chuan Dian Fragment. As shown in Fig. 7A, an amount of rotation is drawn by contour interval of 10°. Strike of each contour is nearly N55°E. From this solution, we conclude that the amount of clockwise rotation systematically increases southeastward from Xichang area in the north to the Chuxiong area in the southeast.

Differential rotation of the Chuan Dian Fragment seems to be recorded in the curvature of the Xianshuihe–Xiaojiang fault system (Fig. 7A). Difference in magnitudes of tectonic rotation between the Xichang and Chuxiong E areas is 44°, which is compatible with the change in curvature of this fault system. The curvature makes a convex shape toward the east. The trend of the Xiaojiang fault system gradually changes up to 40° clockwise toward the south (Leloup et al., 1995; Wang et al., 1998)—is N20°W near Xichang and N20°E on the east side of the Chuxiong basin. Deformation within the Chuan Dian Fragment is thus described by a southeastward increase in tectonic rotation angle as well as the creation of an eastward bulge in the southern part of the Xianshuihe–Xiaojiang fault system.

As a rough estimate, these deformational features are responsible for east–west extension and north–south shortening of the southern part of the Chuan Dian Fragment (Fig. 7B). This deformation was most likely caused by tectonic interaction of the Chuan Dian Fragment with the Yangtze and Indochina blocks. The Chuan Dian Fragment experienced a southward displacement in post-Cretaceous time (Lacassin et al., 1997; Chen et al., 2000; Wang et al., 2001; Tamai et al., 2003). Although statistically insignificant, the present Paleocene–Eocene paleomagnetic data also show its southward displacement ($2.4° \pm 5.1°$) with respect to the Sichuan basin. The Chuan Dian fragment was extruded along the left-lateral Xianshuihe–Xiaojiang fault system under the tectonic influence of India–Asia collision (Avouac and Tapponnier, 1993; Lacassin et al., 1997), but its southward movement was eventually obstructed by the rigid Yangtze and Indochina blocks as larger GPS velocity vectors are observed in the Chuan Dian Fragment than in these blocks (Wang et al., 2001). Due to this obstruction, the studied fragment experienced an east–west extension associated with north–south shortening under the horizontal compressive stresses. As a result of stresses from different blocks, the Chuan Dian Fragment did not behave as a rigid body and underwent an internal deformation.

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